C# & .NET Basic Certification Exam — Paper B

Focus: Arrays, Lists, Memory Allocation (Stack vs Heap) & OOP Fundamentals

Course: Introduction to C# Programming	**Duration:** 60 Minutes
Topic: Basic .NET & C# Fundamentals	**Total Marks:** 50 Marks
Instructions: Write all code legibly. Answer all questions in Sections A, B, and C.	**Passing Grade:** 60%

Section A: Arrays, Lists & Data Collections (15 Marks)

1. (3 Marks) What is the fundamental difference between an array (`int[]`) and a List (`List`) in C#?

A) Arrays can hold strings, while Lists can only hold numbers.
B) Arrays have a fixed size upon creation, whereas Lists are dynamic in size.
C) Arrays store elements on the Heap, whereas Lists store elements on the Stack.
D) Lists do not allow element access via index `list[i]`.

2. (4 Marks) Predict the output of the following C# array processing code:

```
int[] numbers = { 2, 4, 6, 8 };
int sum = 0;
foreach (int num in numbers) {
    sum += num;
}
Console.WriteLine("Average: " + (sum / numbers.Length));
```

3. (8 Marks) Write C# code snippets to: a) Declare and instantiate a dynamic `List` named 'students'. b) Add two names ("Alice" and "Bob") to the list. c) Print the total count of items in the list.

Write your solution below:

Section B: Memory Management: Stack vs. Heap (15 Marks)

4. (5 Marks) In C#, where are value types (such as `int`, `double`, `bool`) typically allocated versus reference types (such as class instances and `string`)?

5. (10 Marks) Analyze the code snippet below and state where each variable and its data are allocated in memory (Stack or Heap):

```
int score = 95;
string studentName = "Maria";
```

Explanation of memory allocation:

Section C: Fundamentals of Object-Oriented Programming (20 Marks)

6. (5 Marks) Define the terms 'Class' and 'Object' in C#, highlighting the relationship between them.

7. (15 Marks) Practical OOP Challenge: Create a C# class named 'Student' with: • Two private fields: `string name` and `int id`. • A constructor that accepts `name` and `id` to initialize the fields. • A public method `DisplayInfo()` that prints the student's name and ID to the console.

Write the complete class definition below:

Exam Paper B — Solutions & Marking Scheme

Question 1 Answer: B

Arrays have a fixed size defined at allocation. Lists (`List`) are dynamic data structures that automatically resize as elements are added or removed.

Question 2 Answer: 'Average: 5'

Explanation: sum = 2 + 4 + 6 + 8 = 20. Array length = 4. Integer division: 20 / 4 = 5.

Question 3 Answer: List Operations Code

List students = new List(); students.Add("Alice"); students.Add("Bob"); Console.WriteLine(students.Count);

Question 4 Answer: Stack vs Heap Allocation

Value types (int, double, bool, struct) are stored directly on the Stack (or inline inside containing objects). Reference types (class objects, strings, arrays) store object references on the Stack while their actual object content resides on the Heap.

Question 5 Answer: Memory Allocation Analysis

• 'score' is a primitive value type (int) stored directly on the Stack holding value 95. • 'studentName' is a reference variable stored on the Stack containing a pointer/reference to the string object "Maria" located on the Heap.

Question 6 Answer: Class vs Object

A Class is a blueprint or template defining fields, properties, and methods. An Object is an instance created from that blueprint residing in memory.

Question 7 Answer: Student Class Solution

public class Student { private string name; private int id; public Student(string name, int id) { this.name = name; this.id = id; } public void DisplayInfo() { Console.WriteLine($"ID: {id}, Name: {name}"); } }